Akoizo Original Practice - Form 2

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) How much energy does a 2.00 kg mass release falling 0.50 m?

A. About 19.6 J

B. About 4.9 J

C. About 9.8 J

D. About 1.0 J

Answer: _____

2. (2 points) When is the Target Distance announced?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. After impound and before the first time slot

B. Weeks before the tournament

C. During the run

D. After both runs

Answer: _____

3. (2 points) If arrival speed doubles with the same stopping distance, how does deceleration change?

A. It quadruples

B. It doubles

C. It halves

D. No change

Answer: _____

4. (2 points) Why add O-rings to wheel rims?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

A. To improve traction and prevent slipping

B. To add mass

C. To slow the vehicle

D. To make wheels bigger

Answer: _____

5. (4 points) Explain why accuracy matters more than speed.

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

Answer: _____
_____
_____

6. (4 points) Why do narrow gaps earn more bonus?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

Answer: _____
_____
_____

7. (4 points) Why include skid distance in the setting?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

Answer: _____
_____
_____

8. (4 points) Compare on-board and launcher designs.

Unit: Energy from a falling mass.
Worked-case data: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?
Separate investigation observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

Answer: _____
_____
_____

9. (4 points) Describe a strategy for the second run after a good first run.

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

Answer: _____
_____
_____

10. (4 points) How can teams practice taping quickly?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?

Answer: _____
_____
_____

12. (2 points) Which vehicle scores better?

Supplied synthetic observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. Vehicle B

B. Vehicle A

C. They tie

Answer: _____

13. (2 points) What explains the short stops?

Supplied synthetic observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

A. The cans slowed the vehicle

B. The egg was too heavy

C. The arc is longer than the straight-line distance

Answer: _____

14. (2 points) Define “Stopping mechanism” in the context of Scrambler.

Answer: _____
_____
_____

15. (2 points) Define “Kinetic energy” in the context of Scrambler.

Answer: _____
_____
_____

16. (2 points) Define “Calibration table” in the context of Scrambler.

Answer: _____
_____
_____

17. (2 points) Define “Deceleration” in the context of Scrambler.

Answer: _____
_____
_____

18. (2 points) Define “Bearing” in the context of Scrambler.

Answer: _____
_____
_____